241 Test 1

PART ONE Instructions: Answer ***FOUR*** of the following counting questions.

1. A test has two parts, 5 questions in part I, and 7 questions in part II. A student has the following options in answering the questions therein. She may answer either 3 of the questions in part I and all of the questions in part II OR all of the questions in part I and 4 of the questions in part II. How many different ways can the student choose a set of questions to answer on this test?

2. How many ways can you form a eight-digit password if the sum of the first and second digits is 16?

3. A group consists of a set of identical triplets and four other people. These people wish to stand in a line for a picture, with the condition that the triplets must stand next to each other. How many ways can these seven people stand and pose for the picture?

4. An office has 10 male workers and 12 female workers. A six-person committee is to be formed. How many such committees are there if at least two women must be chosen?

5. Seven distinct integers from 1-20 (inclusive) are to be used to create a stupid seven-digit password. How many of these passwords have the property that the numbers are in descending order when read from left to right?

PART TWO Instructions: Answer ***TWO*** of the following questions, completing all parts in each. You can answer all three if you would like, and if they are all (mostly) correct, I will give you extra credit. Do not answer more than three, because that means you were probably given the wrong test. Good luck!

1. (a) State Bayes's Rule accurately, in its entirety.
   (b) Prove the following statement: $P(B|A) + P(\bar{B}|A) = 1$

2. A bag contains 3 blue marbles, 2 yellow marbles, and 4 red marbles. A person is to draw two marbles without replacement. If the marbles are the same color, the person wins; if they are different colors, the person loses.

   (a) The person wishes to find determine the probability he/she will win this "game". Construct a sample space S where the probability of picking any element in S is the same.
   (b) What is the probability that this person will win the game?

(c) If it is known that the person won the game, what is the probability that his two marbles were yellow?

3. A lonely teacher owns Pokemon cards from the 1st or 2nd generation, but no others. 80 percent of these Pokemon cards are Snorlax cards. 25 percent of the cards overall are from the 1st generation, and it is known that 60 percent of the 1st generation cards are Snorlax. The teacher randomly draws a card from this set.

   (a) What is the probability that that the teacher draws a 1st generation Snorlax card?

   (b) What is the probability that the teacher draws a 2nd generation Snorlax card?

   (c) If it is known that the teacher draws a 1st generation card, what is the probability that it is a Snorlax?

   (d) If it is known that the teacher draws a 2nd generation card, what is the probability that it is Snorlax *or* not Snorlax?